Huawei Access Node WHI-ACC-50 Failure Recovery Procedure for ALARM_1373: Hardware/Environmental Fault (ASIC Memory Parity Errors, Line Card Crashes, Power Supply Redundancy Loss)

Node: WHI-ACC-50 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_1373

Description

This SOP outlines the recovery procedure for the Huawei Access Node WHI-ACC-50 experiencing a hardware/environmental fault resulting in ASIC memory parity errors, line card crashes, or power supply redundancy loss. The issue is triggered by a failure in the Line Card's ASIC memory, which is a critical component responsible for forwarding packets. The Line Card's ASIC memory is a high-speed memory that stores the routing table, MAC address table, and other forwarding information. When the ASIC memory experiences parity errors, it can lead to line card crashes and power supply redundancy loss. The failure is further exacerbated by the lack of redundancy in the power supply, making it essential to promptly recover the node.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Whitefield East POP to perform physical checks and maintenance.

- Verify the status of the redundant link to ensure it is operational.

- Confirm that the alarm code ALARM_1373 is indeed related to the failure domain of Hardware/Environmental Fault.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display current-configuration, Expected Output: Current configuration of the node, Description: This command will display the current configuration of the node, including the Line Card's ASIC memory settings. This will help identify any configuration issues that may have contributed to the failure.
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Command: display alarm-information, Expected Output: Alarm information for the node, Description: This command will display the alarm information for the node, including the alarm code ALARM_1373. This will confirm that the alarm code is indeed related to the failure domain of Hardware/Environmental Fault.

Command: display system-info, Expected Output: System information for the node, Description: This command will display the system information for the node, including the Line Card's status and power supply redundancy information. This will help identify any issues with the Line Card or power supply.
```

Resolution Steps

Step 1: Step: Replace the failed Line Card, Command: replace line-card {line-card-slot}, Description: Replace the failed Line Card with a new one. This will involve physically replacing the Line Card and then configuring the new Line Card to operate in the same slot.

Step 2: Step: Reconfigure the Line Card's ASIC memory, Command: configuration line-card {line-card-slot} asic-memory, Description: Reconfigure the Line Card's ASIC memory to its default settings. This will involve resetting the ASIC memory to its factory defaults and then re-applying the configuration settings.

Step 3: Step: Verify power supply redundancy, Command: display system-info, Description: Verify that the power supply redundancy is operational. This will involve checking the system information for the node to ensure that the power supply redundancy is online and functioning properly.

Post-Check Verification

```
Command: display current-configuration, Expected Output: Current configuration of the node, Description: This command will verify that the configuration changes have been applied successfully.

Command: display alarm-information, Expected Output: Alarm information for the node, Description: This command will verify that the alarm code ALARM_1373 has been cleared.
```

Roll-back Procedures

- Step: Revert the configuration changes, Command: undo configuration line-card {line-card-slot} asic-memory, Description: Revert the configuration changes made to the Line Card's ASIC memory to its previous settings.

- Step: Revert the Line Card replacement, Command: undo replace line-card {line-card-slot}, Description: Revert the Line Card replacement by removing the new Line Card and re-installing the previous Line Card.